

1	(a)	units of $v^2 = (\text{m s}^{-1})^2 = \text{m s}^{-2}$ units of $u^2 = (\text{m s}^{-1})^2 = \text{m s}^{-2}$ units of $2as = \text{m s}^{-2} \times \text{m} = \text{m s}^{-2}$ Since the units of all the terms are the same, the equation is homogenous. B1 units of v^2 , u^2 , and $2as$ B1 for presentation		
	(b)	(i)	$v^2 = u^2 + 2as$ $a = \frac{v^2}{2s}$ $= \frac{7.7^2}{2(5)}$ $= 5.929$ $= 5.9 \text{ m s}^{-2}$ B1	
		(ii)	$a = \frac{v^2}{2s}$ $\frac{\Delta a}{a} = 2 \frac{\Delta v}{v} + \frac{\Delta s}{s}$ $\Delta a = (5.929) \left(2 \frac{0.3}{7.7} + \frac{0.2}{5.0} \right)$ C1 $= 0.7 \text{ m s}^{-2}$ A1	
		(iii)	$(5.9 \pm 0.7) \text{ m s}^{-2}$	B1
	(c)	(i)	P: Accuracy is defined as how close readings are to their true value. S: Since the <u>average position</u> /point of impact of arrows of Bowman A are <u>closer to the center of the target</u> , compared to bowman B [M1] T: Bowman <u>A</u> is more accurate. [A1]	
		(ii)	P: Precision is defined as how close readings agree with each other. S: Since the arrows of Bowman B are <u>closer to each other</u> compared to bowman A [M1] T: Bowman B is more precise. [A1]	

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2	(a)	By Newton's third law, both vehicles experience the same magnitude of	R1
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